# Decision Log – GCP Marker Detection and Classification

| Decision # | | Reasoning / Rationale | Alternatives Considered | Outcome / Notes |
|---|---|---|---|---|
| 1 | **Model Architecture** – ResNet-18 backbone with two heads (one for keypoint regression, one for shape classification) | ResNet-18 is lightweight yet powerful, pretrained weights help convergence, and two separate heads allow simultaneous regression and classification. | - YOLO / Faster R-CNN for detection- Custom CNN | ResNet-18 worked well with limited compute and small batch sizes; easier to fine-tune. |
| 2 | **Keypoint Prediction (x, y)** | Regression head with MSE loss to predict pixel coordinates. | - Heatmap-based keypoint detection | Regression simpler and works well with fixed-size crops (224x224). Heatmaps more complex for small dataset. |
| 3 | **Shape Classification** | Fully connected layer (512 → 3) with CrossEntropyLoss. | - Use separate CNN classifier | Combined architecture reduces redundancy and allows joint training. |
| 4 | **Image Preprocessing / Transformations** | Resize to 224x224, apply random horizontal flip and small rotations. | - Use original size (2048x1365)- Add color jitter | 224x224 compatible with ResNet input; augmentations help generalization. |
| 5 | **Dataset Handling** | Custom Dataset class that scales coordinates to resized images, only keeps samples with verified shapes. | - Use all images, impute missing shapes | Ensures quality labels for shape classification. |
| 6 | **DataLoader Parameters** | Batch size = 32, shuffle=True, num_workers=2, pin_memory=True | Standard PyTorch best practices for GPU training. | - Larger batch sizes possible on bigger GPU |
| 7 | **Loss Functions** | MSE for keypoints, CrossEntropy for shape, sum as total loss. | Balances regression and classification | - Weighted sum if regression/classification scale differs |
| 8 | **Optimizer and Learning Rate** | Adam, lr=0.0001 | Adam adapts learning rates per parameter, | - SGD with momentum |

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| | | | suitable for small datasets | |
| 9 | **Number of Epochs** | 3 epochs (for demonstration) | Dataset is small, pretrained backbone allows faster convergence | - 10–20 epochs |
| 10 | **Test Inference** | Resize test image to 224x224, predict (x, y) scaled back to original resolution, predict shape. | Consistent with training preprocessing. | - Sliding window over full-size images |
| 11 | **Output Format** | JSON file with structure identical to training labels. | Required for evaluation. | - CSV or other formats |
| 12 | **Evaluation Metrics** | PCK for localization, Macro F1 for classification | Standard metrics for keypoint and class evaluation. | - RMSE for coordinates, accuracy for shape |
| 13 | **Model Saving / Sharing** | torch.save(model.state_dict(), "gcp_model.pth") | Lightweight, easy to reload for inference. | - Save entire model |